Tobias Rubel

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Research Interests: Information Retrieval, Vector Search, Parallel Algorithms, Graph Algorithms, Performance Engineering, Computational Biology

Education

Sep 2021 - Present **PhD Candidate, Computer Science** University of Maryland, College Park, MD.
Advisor: Laxman Dhulipala
Planned Graduation: Aug 2027

Dec. 2019 **B.A. Philosophy**, Reed College, Portland, OR.
Thesis: *Fundamentality and Friends.*
Thesis Advisor: Paul Hovda

Employment

April 2026 - Present	**Student Researcher**, Google Research, College Park, MD.
Aug 2023 - Present	**NSF Graduate Research Fellow**, University of Maryland, College Park, MD.
Jun 2022 - Aug 2023	**Research Assistant**, Molloy Lab at University of Maryland, College Park, MD.
Sep 2021 - Jun 2022	**Teaching Assistant**, University of Maryland, College Park, MD.
May 2018 - Aug 2021	**Research Assistant**, Reed College Center for Advanced Computing, Portland, OR.
Jan 2020 - Aug 2021	**Research Assistant**, Reed College Computational Biology Lab, Portland, OR.

Publications and Talks

Manuscripts in Submission

Tobias Rubel, Richard Wen, Guy Blelloch, Laxman Dhulipala, Lars Gottesbüren, Jakub Łącki, and Vahab Mirrokni. **Turbocharging PiPNN for Proximity and k-NN Graph Building.** Full version. Under submission.

Peer-Reviewed Publications

Tobias Rubel, Richard Wen, Guy Blelloch, Laxman Dhulipala, Lars Gottesbüren, Jakub Łącki, and Vahab Mirrokni. **Fast k-NN Graph Building with Quantized PiPNN and Filtering.** *Similarity Search and Applications (SISAP)* 2026, Springer LNCS. To appear. **First place, Task 1 of the SISAP 2026 Indexing Challenge.** [Code]

Tobias Rubel, Richard Wen, Guy Blelloch, Laxman Dhulipala, Lars Gottesbüren, Jakub Łącki, and Vahab Mirrokni. **Turbocharging PiPNN for Proximity and k-NN Graph Building.** *2nd Workshop on Vector Databases (VecDB) at VLDB* 2026. **Best Paper Award.** [OpenReview]

Tobias Rubel, Richard Wen, Laxman Dhulipala, Lars Gottesbüren, Rajesh Jayaram, and Jakub Łącki. **PiPNN: A Framework for Ultra-Scalable Graph-Based Nearest Neighbor Indexing.** *32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)* 2026. **Best Paper Award, Research Track.** [ACM] [arXiv]

Jamshed Khan, Tobias Rubel, Erin Molloy, Laxman Dhulipala, and Rob Patro. **Fast, parallel, and cache-friendly suffix array construction.** *Algorithms for Molecular Biology* 2024. [AMB]

Junyan Dai, Tobias Rubel, Yunheng Han, and Erin K Molloy. **Leveraging Constraints plus dynamic programming for the large dollo parsimony problem.** *23rd International Workshop on Algorithms in Bioinformatics* 2023. [WABI]

Tobias Rubel*, Ananthan Nambiar*, James McCaull, Jon deVries, and Mark Bedau. **Dropping diversity of products of large US firms: Models and measures.** *Plos One* 2022. [PLOS]

Tobias Rubel*, Pramesh Singh*, and Anna Ritz. **Reconciling signaling pathway databases with network topologies.** *Pacific Symposium on Biocomputing 2022* 2022. [PSB]

Heyuan Zeng, Jinbiao Zhang, Gabriel A Preising, Tobias Rubel, Pramesh Singh, and Anna Ritz. **Graphery: interactive tutorials for biological network algorithms.** *Nucleic Acids Research* 2021. [NAR]

Tobias Rubel, Anna Ritz. **Augmenting Signaling Pathway Reconstructions.** *11th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics* (ACM-BCB) 2020. [ACM]

Mark A. Bedau, Nicholas Gigliotti, Tobias (Rubel) Janssen, Alec Kosik, Ananthan Nambiar, and Norman Packard. **Open-Ended Technological Innovation.** *Artificial Life* 2019. [MIT Press]

Talks

Sep 2026	Ultra-Scalable Graph-Based Nearest Neighbor Indexing with PiPNN. *Meta, FAISS team.* Invited technical talk.
Sep 4, 2026	Turbocharging PiPNN for Proximity and k-NN Graph Building. *2nd Workshop on Vector Databases (VecDB) at VLDB 2026*, Boston, MA.
Aug 2026	Ultra-Scalable Graph-Based Nearest Neighbor Indexing with PiPNN. *KDD 2026*, Jeju Island, Republic of Korea.
Jul 2026	Ultra-Scalable Graph-Based Nearest Neighbor Indexing with PiPNN. *Microsoft.* Invited technical talk.
Nov 6, 2020	Computationally Reconstructing Cellular Signaling Pathways. *Murdock College Science Research Conference 2020*, virtual.
Oct 6, 2020	A Higher-Order Approach to Pathway Reconstruction. *Reed College Computer Science Colloquium*, virtual.
Sep 21, 2020	Augmenting Signaling Pathway Reconstructions. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics 2020*, virtual.

Posters

Jun 25, 2022	Multi-locus species tree estimation with TREE-TMC. *Evolution 2022.*

* Equal Author Contribution.

Software and Technical Skills

Codebases

PiPNN
The parallel C++ implementation of PiPNN, the graph-based nearest neighbor index construction framework from our KDD 2026 paper.
CaPS-SA
The repository of the parallel suffix array algorithm developed jointly by Jamshed Khan, myself, Erin Molloy, Laxman Dhulipala, and Rob Patro.
Programming Languages
C++: I've been working primarily in C++ since 2021.
Python: I've been writing research code in Python since 2018 and have taught the language to many students as a teaching assistant at UMD.

Awards and Funding

Sep 2026	Best Paper Award, 2nd Workshop on Vector Databases (VecDB) at VLDB 2026.
Aug 2026	First Place, Task 1 (k-NN Graph Construction), SISAP 2026 Indexing Challenge.
Aug 2026	Best Paper Award, Research Track, KDD 2026.
Aug 2023	NSF Graduate Research Fellowship. $159,000
Jun 2022	Rita Colwell Travel Fellowship. $800
Jan 2022	PSB Travel Fellowship. $1,500 (declined due to Covid)
Aug 2021	Leidos Fellowship. $10,000